Testing Report

Ala Eh! Facial Recognition Attendance System
Production-hardening release

Generated	14 October 2026, 14:51
Repository state	branch **main**, commit 6d3e382, working tree clean
Machine	Windows build 10.0.26300, Python 3.11.9, Node v24.18.0
Overall result	**ALL AUTOMATED TESTS PASSED**

All numbers in this report come from runs made on the date above. Section 6 states what was **not** tested.

1. Summary

Suite	Tests	Passed	Failed	Skipped	Time
Backend (pytest)	215	215	0	0	38 s
Frontend (vitest)	109	109	0	0	4 s
Total	**324**	**324**	**0**	**0**	

Backend tests were 118 before the hardening work started; 97 were added. Frontend tests were 53 before; 56 were added.

Other checks

Check	Result
TypeScript type check (tsc --noEmit)	**PASS**
Frontend production build	**PASS** ✓ 972 modules transformed. ✓ built in 7.18s
npm audit (production dependencies)	**PASS** found 0 vulnerabilities
Database migrations match the models (fresh database)	**PASS** Database schema is up to date. Models match the migrated schema.
Docker Compose config, production	**PASS**
Docker Compose config, development	**PASS**

2. Backend tests by area

Each row is one test file. Every test runs against a fresh in-memory SQLite database through the real Flask application; recognition engines are replaced by fakes where a real camera would be needed.

Area	File	Tests	Result	What it verifies
Authentication and sessions	test_auth_security.py	49	49/49	Refresh-token cookie with and without CSRF, logout revocation, session cancellation on password / role / deactivation changes, kiosk lock allow-list, KIOSK device role, unlock rate limit, login lockout, archive-vs-delete of employees, audit entries.
Recognition API and enrollment	test_recognition_routes.py	29	29/29	Check-in with blink liveness, cooldown, borrowed-blink handling, enrollment checks, duplicate-face rejection, departments, CSV export, deactivated users.
Shifts, work days and reports	test_shifts_and_reports.py	27	27/27	Overnight 22:00-06:00 shifts as one work day, break across midnight, rotating-shift lateness, stored lateness unaffected by later schedule edits, department audit, future-timestamp rejection, corrections, audit pagination, report grouping.
Guide zone	test_zone.py	26	26/26	Server-side guide-zone check for kiosk frames.
Hardening and security helpers	test_hardening.py	23	23/23	Encryption at rest, input bounds, snapshot handling, anti-spoof modes, image enhancement, Socket.IO origin policy, resilient logging.
Liveness bound to identity	test_liveness_binding.py	17	17/17	Face-track continuity, blink-frame identity check (mismatch drops blinks), multi-frame anti-spoof (worst score), pkg_resources fallback, embedding budget.
Punch rules	test_punch_resolution.py	16	16/16	AM/PM clock-in/out resolution, once-per-day slots, auto lunch-out, shift-end rule, cooldown and re-verification.
Production readiness	test_production_readiness.py	10	10/10	Startup refuses weak or placeholder secrets and SQLite in production, secrets generator output passes the production check, SQLite-to-PostgreSQL copy.
Accuracy tuning tools	test_accuracy_tuning.py	8	8/8	Threshold recommendation and evaluation helpers.
Clear-all-data	test_clear_all.py	8	8/8	Destructive admin action: password check, what is kept (admins, operators, kiosks, audit log).
End-to-end scenarios	test_end_to_end.py	2	2/2	Full employee lifecycle through the API.

3. Frontend tests by area

Component and logic tests run in a simulated browser (jsdom) with the API mocked.

File	Tests	Result	What it verifies
App.test.tsx	10	10/10	Route guards, session restore, kiosk lock and unlock flow, KIOSK device account
services/api.test.ts	11	11/11	Token refresh (single flight), CSRF header, network error vs auth failure, KIOSK_LOCKED event
stores/stores.test.ts	14	14/14	Auth and toast stores
hooks/useCamera.test.ts	5	5/5	Camera on insecure pages, start/stop, frame difference
hooks/usePresenceWake.test.ts	6	6/6	Presence polling and wake-up
hooks/useIdle.test.ts	5	5/5	Idle detection
utils/orgTime.test.ts	7	7/7	Organisation timezone formatting, midnight edge cases, manual-entry timestamps
utils/kioskSessions.test.ts	7	7/7	Kiosk punch/session selection
utils/search.test.ts	6	6/6	Employee search
pages/Attendance/AttendanceKiosk.test.tsx	4	4/4	Single check-in at a time, connection errors, offline pause, wrong unlock password
pages/Records/Records.test.tsx	4	4/4	Manual entry sends a naive timestamp, stale-response guard, paging
pages/Employees/Employees.test.tsx	12	12/12	Single PATCH, KIOSK role, archive wording, clear-all dialog
pages/Review/Review.test.tsx	12	12/12	Review queue: busy guards, spoof snapshots not enrollable, load more
pages/FaceEnrollment/FaceEnrollment.test.tsx	2	2/2	Rejected frames are not resubmitted, calibration commit

4. Live system checks

Run against the real running application (development server with the SQLite database), not against test fakes.

Scenario	Expected	Observed	Result
Sign in as the admin	token issued	OK	**PASS**
Open the user list while the kiosk is locked	403 (blocked)	403	**PASS**
Unlock the kiosk with the password, open the user list	200 (allowed)	200	**PASS**
Live feed (Socket.IO) from a foreign website origin	400 (refused)	400	**PASS**

Docker stack

After the log-folder fix, all five containers came up healthy and the health endpoint through the HTTPS proxy answered **{"database":"ok","ok":true}**. The migrations ran on the existing PostgreSQL database (adopted at the baseline, then upgraded to the hardening revision) without errors.

Container	Status
backend-1	Up 38 minutes (healthy)
caddy-1	Up 38 minutes (healthy)
backup-1	Up 38 minutes
frontend-1	Up 38 minutes (healthy)
db-1	Up 38 minutes (healthy)

Migration scenarios checked during development

Scenario	Result
Brand-new database: upgrade, then models-vs-migrations check	**PASS**
Downgrade to baseline and upgrade again (round trip)	**PASS**
Real legacy database created by the original code, with an existing user: adopted and upgraded, user kept	**PASS**
Development database created by create_all: adopted without errors	**PASS**
Existing PostgreSQL database in Docker: adopted and upgraded	**PASS**

These were run once during the work, not re-run for this report (except the fresh-database check in section 1).

5. Defects found and fixed during testing

Defect	Found by	Fix
Kiosk unlock succeeded but the screen locked itself again, so the admin portal could not be reached	Live use; log showed unlock then lock	Leaving the kiosk is now an explicit state; regression test added (failed before, passes after)
Backend crash-loop in Docker: log folder not writable (root-owned volume from an older setup)	Container logs	Volume recreated; logging now falls back to the console instead of crashing; test added
A borrowed blink still let earlier pooled blink samples count toward liveness	Review of the security workstream	A flagged attempt now wipes the pooled samples; test fails without the fix
Socket.IO login check missing on later app instances in one process	Security workstream tests	Handler carried across instances
Live feed would be refused behind the HTTPS proxy if CORS_ORIGINS missed the site address	Deployment review	Site's own origin always accepted; two tests added
Frontend build failed on a test using Array.at()	tsc / build	Replaced; build and type check clean
Dlib engine and model locks unsafe under the eventlet server; embedding-index rebuild race	Vision workstream code review	OS-level locks, generation counter; tests added

6. What was not tested

Automated tests show the software behaves as specified. They do not show the system is accurate on your cameras or ready for payroll use.

- **Recognition accuracy on real faces.** No false-accept / false-reject rate was measured. Run tools/evaluate_accuracy.py with photos taken at your kiosk before relying on the 0.50 match threshold.
- **Anti-spoofing against real attacks.** The MiniFASNet models are not installed, so passive anti-spoofing only logs. Printed photos and phone screens were not tried. A replayed video containing a blink can pass blink detection.
- **Real kiosk hardware.** No tablet, no HTTPS certificate on a tablet, no Chrome kiosk mode or screen pinning was tried. The camera was not exercised in this run (tests simulate it).
- **Load and performance.** No concurrent-user or burst-rate testing. The design uses one backend worker by design.
- **Browsers.** Tests run in a simulated browser, not in Chrome, Edge, Safari or on a phone.
- **Continuous integration.** The GitHub Actions workflow was written but has not run. The Docker images built and ran locally; their pinned versions were not otherwise verified.
- **Backup and restore.** The backup service is running, but a restore from a backup was not rehearsed.
- **Penetration testing.** Security behaviour is covered by tests, not by an independent attack attempt.
- **Licensing.** Blink detection uses a landmark model with non-commercial licence terms; this is a legal question, not a test.

Appendix A. Every backend test

Accuracy tuning tools (test_accuracy_tuning.py), 8 tests

Test	Result	Time
EngineDefaultsTests: test_kiosk_mirrors_for_sface_only	PASS	1 ms
EngineDefaultsTests: test_sface_threshold_raised_after_full_lfw_evaluation	PASS	1 ms
SFaceMirrorTests: test_one_jitter_is_a_single_embedding	PASS	1 ms
SFaceMirrorTests: test_two_jitters_embeds_original_and_mirror_and_averages	PASS	1 ms
EnrollmentBlurFloorTests: test_enrollment_and_ready_preview_both_use_the_floor	PASS	693 ms
Test	**Result**	**Time**
EnrollmentBlurFloorTests: test_enrollment_floor_is_stricter_than_the_kiosk_floor	PASS	62 ms
EnrollmentBlurFloorTests: test_mid_sharpness_face_passes_kiosk_but_not_enrollment	PASS	40 ms
KioskMirrorTests: test_recognize_burst_embeds_with_the_engine_kiosk_jitters	PASS	43 ms

Authentication and sessions (test_auth_security.py), 49 tests

Test	Result	Time
LoginAndRefreshTests: test_access_token_cannot_be_used_to_refresh	PASS	307 ms
LoginAndRefreshTests: test_bearer_refresh_token_works_without_cookies	PASS	309 ms
LoginAndRefreshTests: test_cookie_refresh_requires_the_csrf_header	PASS	318 ms
LoginAndRefreshTests: test_cookie_wins_over_an_access_token_in_the_authorization_header	PASS	312 ms
LoginAndRefreshTests: test_deactivation_is_only_revealed_after_the_right_password	PASS	587 ms
Test	**Result**	**Time**
LoginAndRefreshTests: test_failed_login_is_logged_with_email_and_ip	PASS	303 ms
LoginAndRefreshTests: test_login_puts_refresh_token_only_in_an_httponly_cookie	PASS	305 ms
LoginAndRefreshTests: test_unknown_email_costs_a_bcrypt_check_and_gives_the_same_error	PASS	303 ms
LogoutAndRevocationTests: test_deactivation_revokes_tokens	PASS	334 ms
LogoutAndRevocationTests: test_expired_revocations_are_purged	PASS	305 ms
LogoutAndRevocationTests: test_logout_never_fails	PASS	37 ms
LogoutAndRevocationTests: test_logout_revokes_refresh_and_access_tokens_and_clears_cookies	PASS	308 ms
LogoutAndRevocationTests: test_password_change_signs_the_user_out_everywhere	PASS	867 ms
LogoutAndRevocationTests: test_profile_edit_does_not_sign_the_user_out	PASS	317 ms
LogoutAndRevocationTests: test_revoking_sign_in_revokes_tokens	PASS	322 ms
LogoutAndRevocationTests: test_role_change_revokes_tokens	PASS	323 ms
KioskLockTests: test_kiosk_refresh_token_lives_longer	PASS	311 ms
KioskLockTests: test_lock_revokes_the_unlocked_session_and_is_audited	PASS	317 ms
KioskLockTests: test_locked_session_only_reaches_the_kiosk_allowlist	PASS	339 ms
KioskLockTests: test_portal_session_cannot_check_in	PASS	103 ms
KioskLockTests: test_refresh_keeps_the_kiosk_claim	PASS	327 ms
KioskLockTests: test_staff_cannot_lock_or_check_in	PASS	34 ms
KioskLockTests: test_unlock_is_rate_limited	PASS	1728 ms
KioskLockTests: test_unlock_with_the_password_restores_the_portal	PASS	605 ms
KioskLockTests: test_unlock_with_the_wrong_password_is_refused_and_audited	PASS	599 ms
KioskRoleTests: test_admin_kiosk_device_id_is_prefixed_with_the_account	PASS	37 ms

Test	Result	Time
KioskRoleTests: test_kiosk_account_is_locked_even_with_a_token_that_says_otherwise	PASS	32 ms
KioskRoleTests: test_kiosk_account_login_is_always_locked_and_cannot_unlock	PASS	310 ms
KioskRoleTests: test_kiosk_accounts_cannot_be_enrolled	PASS	39 ms
KioskRoleTests: test_kiosk_accounts_need_a_password	PASS	310 ms
KioskRoleTests: test_kiosk_device_id_is_the_account_not_the_client_value	PASS	37 ms
KioskRoleTests: test_kiosk_sessions_get_no_live_feed	PASS	307 ms
RateLimitTests: test_account_locks_after_repeated_failures_even_with_the_right_password	PASS	1674 ms
RateLimitTests: test_limits_can_be_disabled	PASS	3632 ms
RateLimitTests: test_lockout_expires	PASS	26 ms
RateLimitTests: test_lockout_is_case_insensitive	PASS	1405 ms
RateLimitTests: test_login_is_limited_per_ip	PASS	2784 ms
RateLimitTests: test_sliding_window_frees_slots_as_they_age	PASS	26 ms
RateLimitTests: test_success_resets_the_failure_count	PASS	2814 ms
UserAdminTests: test_creation_is_audited_without_the_password	PASS	308 ms
UserAdminTests: test_list_is_paginated_with_a_total_header	PASS	121 ms
UserAdminTests: test_login_roles_cannot_lose_sign_in	PASS	31 ms
UserAdminTests: test_orm_delete_never_cascades_into_attendance	PASS	29 ms
UserAdminTests: test_overlong_password_is_rejected	PASS	31 ms
UserAdminTests: test_updates_are_audited_without_password_values	PASS	323 ms
UserAdminTests: test_user_with_attendance_is_archived_not_deleted	PASS	44 ms
UserAdminTests: test_user_without_attendance_is_deleted	PASS	34 ms
SpoofSnapshotTests: test_snapshot_cannot_be_added_twice	PASS	34 ms
SpoofSnapshotTests: test_spoof_snapshot_cannot_become_a_face_sample	PASS	33 ms

Clear-all-data (test_clear_all.py), 8 tests

Test	Result	Time
ClearAllDataTests: test_admin_can_still_sign_in_afterwards	PASS	586 ms
ClearAllDataTests: test_audit_history_is_kept_and_the_reset_itself_is_recorded	PASS	316 ms
ClearAllDataTests: test_missing_confirmation_deletes_nothing	PASS	44 ms
ClearAllDataTests: test_non_json_body_is_a_clean_error	PASS	36 ms
ClearAllDataTests: test_only_administrators_may_clear	PASS	39 ms
Test	Result	Time
ClearAllDataTests: test_operators_can_be_removed_too_on_request	PASS	308 ms
ClearAllDataTests: test_wipes_company_data_but_keeps_admins_and_departments	PASS	325 ms
ClearAllDataTests: test_wrong_password_deletes_nothing	PASS	311 ms

End-to-end scenarios (test_end_to_end.py), 2 tests

Test	Result	Time
EndToEndTests: test_full_attendance_journey	PASS	1722 ms
EndToEndTests: test_staff_role_is_fenced_off_from_admin_features	PASS	36 ms

Hardening and security helpers (test_hardening.py), 23 tests

Test	Result	Time
EnhanceTests: test_disabled_flag_skips_enhancement	PASS	1 ms
EnhanceTests: test_evenly_lit_textured_face_is_left_alone	PASS	2 ms
EnhanceTests: test_side_lit_face_is_normalised	PASS	2 ms
JitterTests: test_variants_cover_requested_jitters	PASS	1 ms
AntiSpoofTests: test_flat_print_scores_low	PASS	1 ms
Test	**Result**	**Time**
AntiSpoofTests: test_mode_resolution	PASS	0 ms
AntiSpoofTests: test_natural_texture_scores_high	PASS	8 ms
AntiSpoofTests: test_pixel_grid_moire_scores_low	PASS	3 ms
AccuracyEvalTests: test_curve_separates_known_from_unknown	PASS	2 ms
CryptoTests: test_embeddings_encrypted_at_rest_but_still_match	PASS	40 ms
CryptoTests: test_plaintext_when_no_key	PASS	26 ms
CryptoTests: test_roundtrip_and_legacy_plaintext_still_readable	PASS	27 ms
CryptoTests: test_wrong_key_raises	PASS	29 ms
ConsentAndAuditTests: test_enrollment_requires_consent_then_remembers_it	PASS	61 ms
SnapshotReviewTests: test_enroll_from_snapshot_adds_sample_and_resolves	PASS	58 ms
SnapshotReviewTests: test_matched_checkin_keeps_audit_snapshot_linked_to_event	PASS	53 ms
SnapshotReviewTests: test_old_snapshots_are_purged	PASS	27 ms
SnapshotReviewTests: test_review_actions_and_admin_only	PASS	52 ms
SnapshotReviewTests: test_spoof_suspected_blocks_punch_and_is_queued	PASS	36 ms
SnapshotReviewTests: test_unknown_scan_lands_in_pending_queue_and_is_throttled	PASS	55 ms
SocketOriginTests: test_forwarded_host_ignored_without_trusted_proxy	PASS	1 ms
SocketOriginTests: test_same_origin_behind_trusted_proxy	PASS	1 ms
LoggingResilienceTests: test_unwritable_log_dir_does_not_crash_startup	PASS	2 ms

Liveness bound to identity (test_liveness_binding.py), 17 tests

Test	Result	Time
TrackContinuityTests: test_alternating_faces_never_pool_the_other_faces_blink	PASS	45 ms
TrackContinuityTests: test_ears_only_come_from_the_identity_track	PASS	48 ms
TrackContinuityTests: test_one_continuous_track_keeps_every_sample	PASS	139 ms
TrackContinuityTests: test_track_helpers	PASS	23 ms
BlinkFrameIdentityTests: test_adjacent_open_eye_frame_of_someone_else_drops_the_ears	PASS	44 ms
Test	**Result**	**Time**
BlinkFrameIdentityTests: test_blink_candidate_needs_a_clear_drop	PASS	25 ms
BlinkFrameIdentityTests: test_check_can_be_disabled	PASS	43 ms
BlinkFrameIdentityTests: test_closed_eye_frame_of_someone_else_drops_the_ears	PASS	50 ms
BlinkFrameIdentityTests: test_closed_eye_frame_of_the_same_person_passes	PASS	47 ms
BlinkFrameIdentityTests: test_no_blink_candidate_costs_no_extra_embeddings	PASS	44 ms
BlinkFrameIdentityTests: test_typical_ten_frame_burst_stays_cheap	PASS	52 ms
MultiFrameSpoofTests: test_all_real_frames_pass	PASS	36 ms
MultiFrameSpoofTests: test_frame_count_is_configurable	PASS	39 ms
MultiFrameSpoofTests: test_most_suspicious_embedded_frame_wins	PASS	37 ms

Test	Result	Time
PkgResourcesCompatTests: test_face_recognition_models_resolve	PASS	1 ms
PkgResourcesCompatTests: test_installs_only_when_the_real_module_is_missing	PASS	1 ms
PkgResourcesCompatTests: test_no_top_level_shadow_module	PASS	1 ms

Production readiness (test_production_readiness.py), 10 tests

Test	Result	Time
ConfigCheckTests: test_development_only_warns	PASS	18 ms
ConfigCheckTests: test_good_config_has_no_problems	PASS	1 ms
ConfigCheckTests: test_production_refuses_unsafe_config	PASS	2 ms
ConfigCheckTests: test_production_starts_with_safe_config	PASS	2 ms
ConfigCheckTests: test_shipped_defaults_are_flagged	PASS	0 ms
Test	**Result**	**Time**
ConfigCheckTests: test_short_identical_and_invalid_values_are_flagged	PASS	1 ms
GenerateSecretsTests: test_generated_values_pass_the_production_check	PASS	1 ms
GenerateSecretsTests: test_template_renders_every_placeholder	PASS	0 ms
GenerateSecretsTests: test_values_are_unique_per_run	PASS	1 ms
MigrationTests: test_copies_every_row_and_refuses_a_non_empty_target	PASS	139 ms

Punch rules (test_punch_resolution.py), 16 tests

Test	Result	Time
PunchResolutionTests: test_am_out_pressed_before_timing_in_records_the_time_in	PASS	29 ms
PunchResolutionTests: test_am_selection_in_the_afternoon_is_moved_to_pm	PASS	38 ms
PunchResolutionTests: test_auto_selection_needs_no_correction	PASS	29 ms
PunchResolutionTests: test_lunch_break_not_noon_splits_am_from_pm	PASS	41 ms
PunchResolutionTests: test_manual_admin_entry_is_written_exactly_as_asked	PASS	38 ms
Test	**Result**	**Time**
PunchResolutionTests: test_no_regular_time_in_once_the_shift_has_ended	PASS	47 ms
PunchResolutionTests: test_normal_day_fills_the_four_slots_in_order	PASS	156 ms
PunchResolutionTests: test_overtime_still_recordable_after_the_shift_ends	PASS	53 ms
PunchResolutionTests: test_pm_in_at_end_of_shift_is_recorded_as_pm_out	PASS	40 ms
PunchResolutionTests: test_regular_chain_spans_am_and_pm	PASS	31 ms
PunchResolutionTests: test_rejects_an_unknown_session	PASS	27 ms
PunchResolutionTests: test_repeating_a_punch_after_the_window_does_not_duplicate_it	PASS	52 ms
PunchResolutionTests: test_rescan_inside_the_window_verifies_instead_of_recording	PASS	40 ms
PunchResolutionTests: test_rescan_within_the_cooldown_is_reported_separately	PASS	35 ms
PunchResolutionTests: test_shifting_department_has_no_shift_end_rule	PASS	41 ms
PunchResolutionTests: test_yesterdays_events_do_not_carry_over	PASS	30 ms

Recognition API and enrollment (test_recognition_routes.py), 29 tests

Test	Result	Time
MatcherTests: test_classify_statuses	PASS	1 ms
MatcherTests: test_rank_users_combines_max_and_centroid	PASS	1 ms
LivenessTests: test_blink_with_both_eyes_passes	PASS	1 ms
LivenessTests: test_one_eye_only_fails	PASS	1 ms
LivenessTests: test_photo_landmark_noise_fails	PASS	1 ms
Test	**Result**	**Time**
LivenessTests: test_tracker_pools_samples_across_scans	PASS	1 ms
EmbeddingIndexTests: test_deactivated_users_are_not_matched	PASS	34 ms
EmbeddingIndexTests: test_index_ranks_enrolled_user_first	PASS	25 ms
CheckInRouteTests: test_blank_frames_return_no_face	PASS	51 ms
CheckInRouteTests: test_borrowed_blink_discards_pooled_samples	PASS	49 ms
CheckInRouteTests: test_first_scan_of_new_day_is_clock_in	PASS	30 ms
CheckInRouteTests: test_match_with_blink_records_clock_in_then_cooldown	PASS	56 ms
CheckInRouteTests: test_match_without_blink_is_not_recorded	PASS	37 ms
CheckInRouteTests: test_requires_authentication	PASS	32 ms
EmployeeAndEnrollmentTests: test_admin_creates_employee_without_login	PASS	303 ms
EmployeeAndEnrollmentTests: test_duplicate_employee_code_rejected	PASS	29 ms
EmployeeAndEnrollmentTests: test_enroll_accepts_consistent_sample_and_counts_it	PASS	201 ms
EmployeeAndEnrollmentTests: test_enroll_rejects_face_already_enrolled_as_someone_else	PASS	43 ms
EmployeeAndEnrollmentTests: test_enroll_rejects_inconsistent_face	PASS	41 ms
EmployeeAndEnrollmentTests: test_legacy_samples_from_other_engine_are_reported_outdated	PASS	35 ms
EmployeeAndEnrollmentTests: test_photo_endpoint_serves_enrolled_photo	PASS	44 ms
EmployeeAndEnrollmentTests: test_public_registration_disabled	PASS	27 ms
EmployeeAndEnrollmentTests: test_reset_only_affects_target	PASS	35 ms
EmployeeAndEnrollmentTests: test_staff_cannot_create_or_enroll	PASS	36 ms
SessionAndDepartmentValidationTests: test_deactivated_user_session_is_unauthorized	PASS	34 ms
SessionAndDepartmentValidationTests: test_invalid_shifting_value_does_not_clear_department_shifts	PASS	38 ms
SessionAndDepartmentValidationTests: test_revoking_login_invalidates_existing_access_and_refresh_tokens	PASS	36 ms
ReportTests: test_export_includes_session_and_escapes_formula_text	PASS	41 ms
ReportTests: test_today_and_csv	PASS	53 ms

Shifts, work days and reports (test_shifts_and_reports.py), 27 tests

Test	Result	Time
OvernightFixedShiftTests: test_auto_lunch_out_across_midnight	PASS	40 ms
OvernightFixedShiftTests: test_break_punches_split_halves_after_midnight	PASS	65 ms
OvernightFixedShiftTests: test_in_before_and_out_after_midnight_is_one_work_day	PASS	50 ms
OvernightFixedShiftTests: test_late_and_early_leave_across_midnight	PASS	44 ms
OvernightFixedShiftTests: test_shift_over_next_morning_is_a_time_out_not_a_new_day	PASS	174 ms
Test	**Result**	**Time**
OvernightFixedShiftTests: test_today_report_counts_night_worker_currently_in_after_midnight	PASS	44 ms

Test	Result	Time
RotatingShiftTests: test_early_arrival_matches_upcoming_shift_and_is_not_late	PASS	36 ms
RotatingShiftTests: test_graveyard_shift_pairs_across_midnight	PASS	53 ms
RotatingShiftTests: test_late_for_afternoon_shift	PASS	36 ms
RotatingShiftTests: test_no_matching_shift_is_never_late	PASS	33 ms
RotatingShiftTests: test_on_time_for_afternoon_shift	PASS	35 ms
StampedLatenessTests: test_editing_the_department_does_not_rewrite_history	PASS	54 ms
StampedLatenessTests: test_legacy_rows_fall_back_to_the_current_schedule	PASS	39 ms
DepartmentAuditTests: test_create_update_delete_are_audited	PASS	37 ms
DepartmentAuditTests: test_no_op_update_is_not_audited	PASS	31 ms
DepartmentAuditTests: test_schedule_validation	PASS	51 ms
ManualEntryAndCorrectionTests: test_bad_timestamp_rejected	PASS	29 ms
ManualEntryAndCorrectionTests: test_correction_changes_type_and_session	PASS	46 ms
ManualEntryAndCorrectionTests: test_correction_rejects_future_and_bad_values	PASS	48 ms
ManualEntryAndCorrectionTests: test_correction_with_timestamp_change_restamps_and_audits	PASS	51 ms
ManualEntryAndCorrectionTests: test_deleting_the_first_time_in_moves_lateness_to_the_next	PASS	160 ms
ManualEntryAndCorrectionTests: test_future_timestamp_rejected	PASS	39 ms
ManualEntryAndCorrectionTests: test_naive_timestamp_is_org_local_and_session_follows_the_break	PASS	54 ms
PaginationTests: test_attendance_offset_and_total_header	PASS	46 ms
PaginationTests: test_audit_paging_and_filters	PASS	45 ms
ReportGroupingTests: test_late_flag_ignores_an_ot_time_in_before_the_shift	PASS	49 ms
ReportGroupingTests: test_rows_group_by_work_date_and_csv_carries_it	PASS	94 ms

Guide zone (test_zone.py), 26 tests

Test	Result	Time
ParseTests: test_missing_or_bad_input_means_no_zone	PASS	1 ms
ParseTests: test_valid_zone	PASS	1 ms
GeometryTests: test_background_face_is_filtered_out	PASS	1 ms
GeometryTests: test_centre_is_inside	PASS	0 ms
GeometryTests: test_far_to_the_side_is_outside	PASS	1 ms
Test	Result	Time
GeometryTests: test_slightly_off_centre_still_scans	PASS	0 ms
GeometryTests: test_small_face_is_too_small	PASS	1 ms
CheckInWiringTests: test_no_zone_means_no_restriction	PASS	43 ms
CheckInWiringTests: test_zone_is_passed_to_the_scanner	PASS	31 ms
EnrollmentZoneTests: test_face_inside_the_circle_enrols	PASS	43 ms
EnrollmentZoneTests: test_face_outside_the_circle_is_refused	PASS	34 ms
EnrollmentZoneTests: test_live_check_tells_the_admin_to_move	PASS	33 ms
EnrollmentZoneTests: test_no_zone_keeps_the_old_behaviour	PASS	42 ms
EnrollmentZoneTests: test_passer_by_outside_the_circle_is_ignored	PASS	55 ms
EnrollmentZoneTests: test_tiny_face_is_asked_to_come_closer	PASS	33 ms
EnrollmentZoneTests: test_unknown_pose_is_rejected	PASS	33 ms

Test	Result	Time
KioskConfigTests: test_requires_a_signed_in_user	**PASS**	25 ms
KioskConfigTests: test_returns_org_time_and_the_lunch_break	**PASS**	30 ms
PresenceTests: test_a_close_person_counts_even_with_a_passer_by	**PASS**	30 ms
PresenceTests: test_a_frame_is_required_and_login_too	**PASS**	32 ms
PresenceTests: test_does_not_store_anything	**PASS**	33 ms
PresenceTests: test_nobody_in_frame	**PASS**	34 ms
PresenceTests: test_passer_by_outside_the_circle_does_not_count	**PASS**	171 ms
PresenceTests: test_someone_far_back_does_not_count	**PASS**	36 ms
PresenceTests: test_someone_in_the_circle_is_present	**PASS**	32 ms
PresenceTests: test_without_a_zone_any_face_counts	**PASS**	34 ms

Appendix B. Every frontend test

App.test.tsx, 10 tests

Test	Result	Time
route guards > sends a signed-out visitor to the sign-in page	PASS	64 ms
route guards > restores the session from the refresh cookie without flashing the sign-in page	PASS	44 ms
route guards > shows a retry screen, not the sign-in page, while the server is unreachable	PASS	14 ms
route guards > lets an admin open the review page	PASS	35 ms
route guards > keeps staff out of admin pages and sends them to their own attendance	PASS	14 ms
Test	**Result**	**Time**
route guards > never shows the kiosk to staff	PASS	14 ms
route guards > keeps every route on the kiosk while the session is kiosk-locked	PASS	112 ms
route guards > locks the session on the server when the kiosk opens	PASS	28 ms
route guards > leaves the kiosk for the admin portal after the password is accepted, without re-locking	PASS	713 ms
route guards > sends a KIOSK device account straight to the kiosk, with no way out	PASS	31 ms

hooks/useCamera.test.ts, 5 tests

Test	Result	Time
useCamera > explains that the camera needs HTTPS on an insecure page	PASS	22 ms
useCamera > reports the same when the browser has no camera API	PASS	3 ms
useCamera > starts the camera when it is available	PASS	6 ms
frameDifference > is 0 for identical frames and grows with change	PASS	2 ms
frameDifference > is Infinity when frames can't be compared	PASS	0 ms

hooks/useIdle.test.ts, 5 tests

Test	Result	Time
useIdle > goes idle after the timeout and wakes on activity	PASS	26 ms
useIdle > wakes on the wake event, with no touch at all	PASS	4 ms
useIdle > activity restarts the countdown	PASS	3 ms
useIdle > never goes idle while paused, and restarts the full countdown when resumed	PASS	3 ms
useIdle > pausing while idle wakes it (a scan starting must clear the idle screen)	PASS	3 ms

hooks/usePresenceWake.test.ts, 6 tests

Test	Result	Time
usePresenceWake > wakes for someone who arrives after the spot was empty	PASS	22 ms
usePresenceWake > needs consecutive sightings: one noisy frame does not wake it	PASS	3 ms
usePresenceWake > ignores someone who was already standing there when the idle screen came up	PASS	3 ms
usePresenceWake > wakes for that person once they have stepped away and come back	PASS	3 ms
usePresenceWake > does nothing while disabled, and stops when disabled again	PASS	5 ms
Test	**Result**	**Time**
usePresenceWake > treats a failed check as no information, neither waking nor counting as empty	PASS	3 ms

services/api.test.ts, 11 tests

Test	Result	Time
api client > talks to the same-origin /api with cookies enabled	PASS	12 ms
api client > sends the in-memory access token, and never stores it	PASS	5 ms
api client > refreshes with the CSRF header and the cookie, not a bearer token	PASS	5 ms
api client > refreshes once for concurrent 401s and retries each request with the new token	PASS	28 ms
api client > treats TOKEN_REVOKED like expiry: one refresh, then the retry	PASS	4 ms
Test	**Result**	**Time**
api client > ends the session when the refresh itself is rejected (401)	PASS	11 ms
api client > keeps the session when the refresh fails on the network (a Wi-Fi blip)	PASS	4 ms
api client > keeps the session when the refresh gets a 5xx	PASS	3 ms
api client > does not refresh-and-retry a failed login	PASS	2 ms
api client > announces a kiosk-locked session (403 KIOSK_LOCKED)	PASS	2 ms
api client > explains 429s with the server's retry time	PASS	1 ms

stores/stores.test.ts, 14 tests

Test	Result	Time
toastStore > queues a toast and removes it after its timeout	PASS	7 ms
toastStore > keeps error toasts on screen longer than success toasts	PASS	1 ms
toastStore > shows at most four toasts, dropping the oldest	PASS	1 ms
authStore > keeps the access token in memory only; nothing secret goes to storage	PASS	9 ms
authStore > follows the server's kiosk lock	PASS	1 ms
Test	**Result**	**Time**
authStore > revokes the session on the server, then forgets it	PASS	7 ms
authStore > still signs out locally when the server can't be reached	PASS	7 ms
authStore > restores the session from the refresh cookie on load	PASS	2 ms
authStore > skips the restore round-trip when there is no session cookie	PASS	1 ms
authStore > goes to the sign-in page when the refresh cookie is rejected	PASS	2 ms
authStore > reports offline (not signed out) when the server is unreachable	PASS	1 ms
authStore > locks and unlocks the kiosk through the server	PASS	2 ms
authStore > rejects a wrong unlock password and stays locked	PASS	3 ms
authStore > drops the session when the API layer reports it expired	PASS	1 ms

utils/guideZone.test.ts, 4 tests

Test	Result	Time
guideZone > maps a centred oval to the centre of the frame	PASS	3 ms
guideZone > mirrors left and right: an oval on the screen's left is on the frame's right	PASS	1 ms
guideZone > accounts for the part of the frame cropped off by object-cover	PASS	0 ms
guideZone > returns null when it cannot measure	PASS	0 ms

utils/kioskSessions.test.ts, 7 tests

Test	Result	Time
kiosk morning / afternoon rules > does not offer Afternoon before the afternoon starts	PASS	2 ms
kiosk morning / afternoon rules > opens Afternoon time in when lunch starts, but not Afternoon time out yet	PASS	0 ms
kiosk morning / afternoon rules > after lunch only Afternoon is offered	PASS	0 ms
kiosk morning / afternoon rules > never restricts overtime	PASS	0 ms
kiosk morning / afternoon rules > falls back to noon until the server's break window is known	PASS	0 ms
Test	**Result**	**Time**
kiosk morning / afternoon rules > picks the sensible default for the time of day	PASS	0 ms
kiosk morning / afternoon rules > explains why a choice is unavailable	PASS	0 ms

utils/orgTime.test.ts, 7 tests

Test	Result	Time
orgTime > formats instants in the organisation's timezone, not the browser's	PASS	5 ms
orgTime > uses the timezone from /api/meta once it is loaded	PASS	3 ms
orgTime > ignores an unknown timezone	PASS	1 ms
orgTime > corrects a wrong tablet clock with the server's	PASS	1 ms
orgTime > works out 'today' across midnight per timezone	PASS	6 ms
Test	**Result**	**Time**
orgTime > formats a bare calendar date as that date in every timezone	PASS	1 ms
orgTime > sends datetime-local values as naive org-local wall time	PASS	1 ms

utils/search.test.ts, 6 tests

Test	Result	Time
smartMatch > matches everything on an empty query	PASS	2 ms
smartMatch > finds words across different fields in any order	PASS	1 ms
smartMatch > is case-insensitive	PASS	0 ms
smartMatch > requires every word to match	PASS	0 ms
smartMatch > does not match a word split across two fields	PASS	0 ms
Test	**Result**	**Time**
smartMatch > ignores null and undefined fields	PASS	0 ms

pages/Attendance/AttendanceKiosk.test.tsx, 4 tests

Test	Result	Time
AttendanceKiosk > sends one check-in at a time and locks the punch choice while it runs	PASS	549 ms
AttendanceKiosk > says plainly that a check-in was not recorded when the network fails	PASS	80 ms
AttendanceKiosk > pauses scanning while offline	PASS	30 ms
AttendanceKiosk > unlocks through the server and reports a wrong password	PASS	220 ms

pages/FaceEnrollment/FaceEnrollment.test.tsx, 2 tests

Test	Result	Time
Face enrollment capture > does not resubmit a rejected frame and does not calibrate from it	PASS	3292 ms
Face enrollment capture > commits calibration only after the sample is accepted	PASS	736 ms

pages/Records/Records.test.tsx, 4 tests

Test	Result	Time
Records page > sends a manual entry as naive org-local wall time (no browser timezone conversion)	PASS	469 ms
Records page > asks for an employee and a reason before saving a manual entry	PASS	119 ms
Records page > ignores a slow response for an older filter	PASS	132 ms
Records page > pages the scan log with limit/offset and the total from X-Total-Count	PASS	120 ms

pages/Employees/Employees.test.tsx, 12 tests

Test	Result	Time
Employees page > lists employees with their face-enrollment state	PASS	107 ms
Employees page > filters with the smart search	PASS	158 ms
Employees page > clears all data only after the password and DELETE are entered	PASS	446 ms
Employees page > can also remove operator accounts when asked	PASS	512 ms
Employees page > shows the server's message and keeps the data when the password is wrong	PASS	349 ms
Test	**Result**	**Time**
Employees page > keeps the data when the confirmation is cancelled	PASS	167 ms
Employees page > hides the clear-all button from non-admins	PASS	32 ms
Employees page > asks before deleting an employee and explains archiving	PASS	172 ms
Employees page > keeps an archived employee in the list as deactivated and says so	PASS	148 ms
Employees page > saves a role change and its password in ONE request	PASS	304 ms
Employees page > requires a password before turning a staff member into a kiosk device account	PASS	179 ms
Employees page > pages a long employee list	PASS	217 ms

pages/Review/Review.test.tsx, 12 tests

Test	Result	Time
Review page > lists pending scans with the closest employee and score	PASS	72 ms
Review page > shows an empty state when nothing needs review	PASS	15 ms
Review page > switches filters and reloads	PASS	151 ms
Review page > dismisses a snapshot	PASS	67 ms
Review page > asks for confirmation before deleting a snapshot	PASS	88 ms
Test	**Result**	**Time**
Review page > adds a frame as a sample for the preselected closest employee	PASS	113 ms
Review page > asks the admin to confirm consent when none is on file, then retries with it	PASS	129 ms
Review page > offers no review actions on a matched check-in	PASS	21 ms
Review page > never offers to enroll a suspected spoof	PASS	17 ms
Review page > explains when the server refuses a frame (409) instead of a generic error	PASS	79 ms
Review page > sends a dismiss only once on a double click	PASS	102 ms
Review page > loads more snapshots from the next offset	PASS	97 ms